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$$\begin{aligned} I &= \int e^x \sin 5x dx \\ &= e^x \sin 5x - 5 \int e^x \cos 5x dx \\ &= e^x \sin 5x - 5 \left(e^x \cos 5x + 5 \int e^x \sin 5x dx \right) \\ &= e^x \sin 5x - 5e^x \cos 5x - 25I + C \\ \Rightarrow I &= \frac{e^x \sin 5x - 5e^x \cos 5x}{26} + C \end{aligned}$$

References